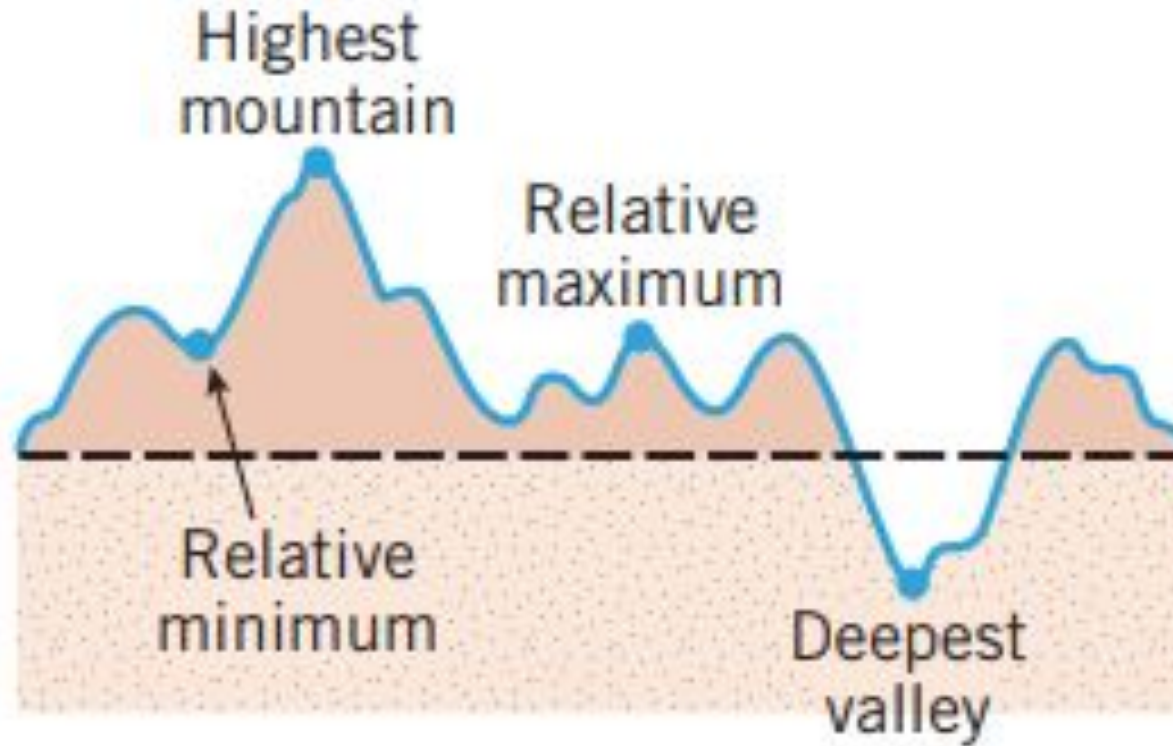


MAXIMA AND MINIMA , OPTIMIZATION

Prepared by
Md Saif Kabir
Lecturer,OAA

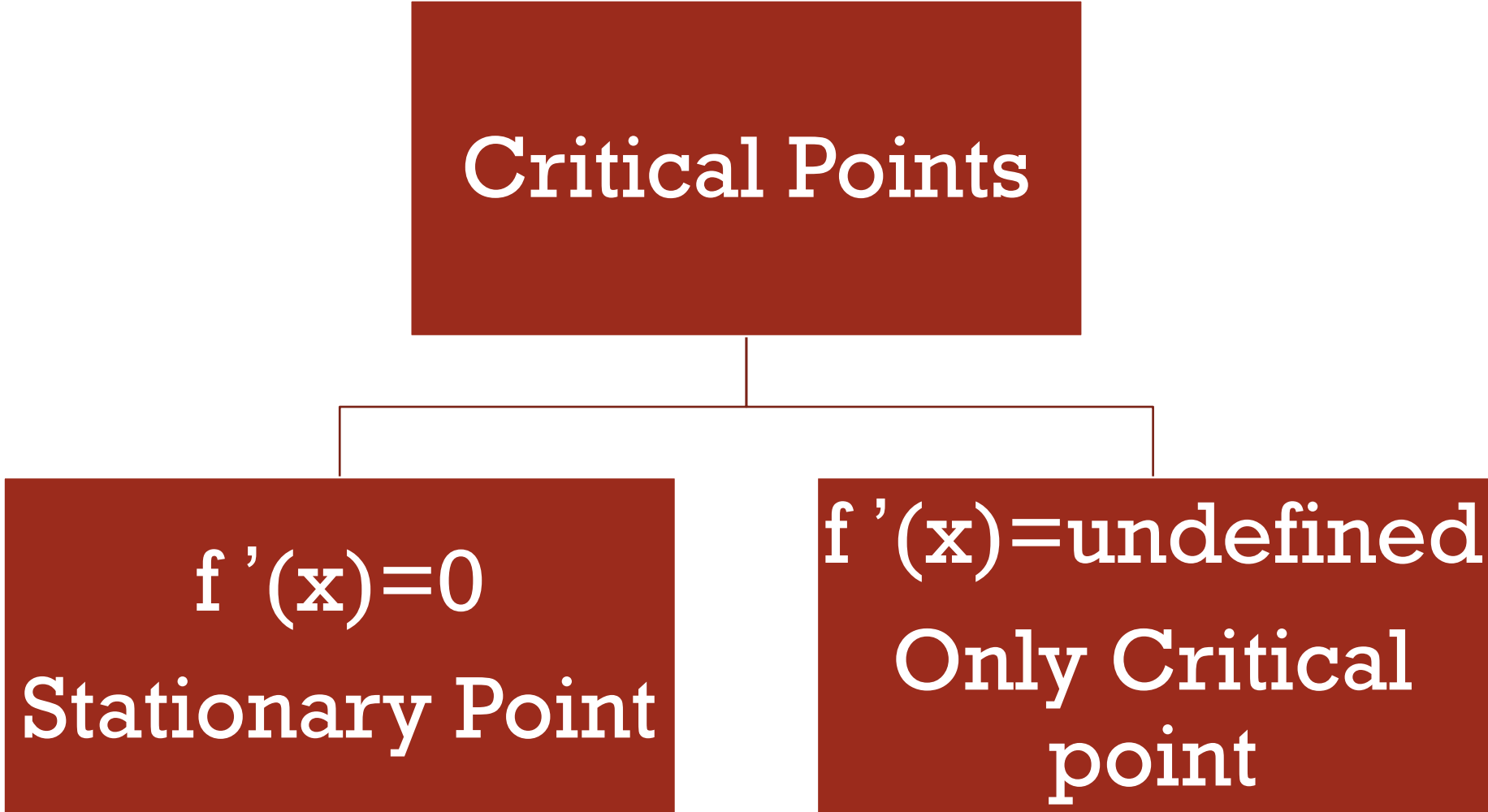
RELATIVE MAXIMA AND MINIMA



▲ Figure 4.2.1



Critical Points



```
graph TD; A[Critical Points] --> B["f'(x)=0<br/>Stationary Point"]; A --> C["f'(x)=undefined<br/>Only Critical point"]
```

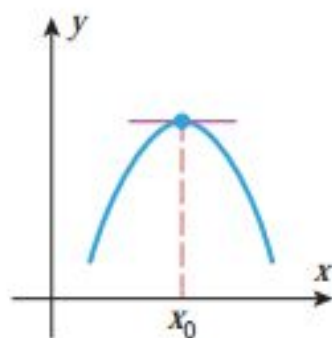
$$f'(x)=0$$

Stationary Point

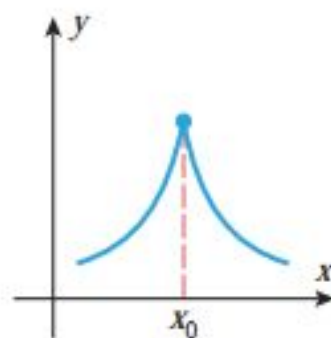
$$f'(x)=\text{undefined}$$

Only Critical
point

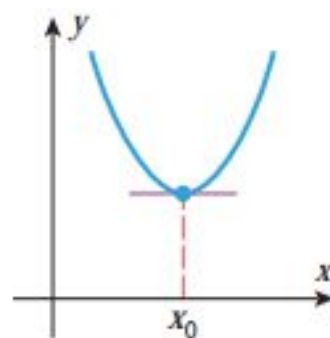




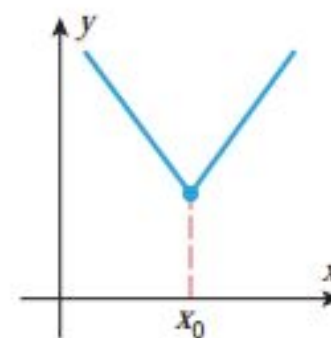
Critical point
Stationary point
Relative maximum



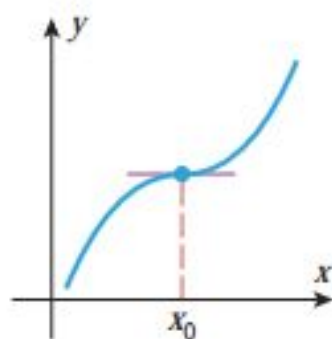
Critical point
Not a stationary point
Relative maximum



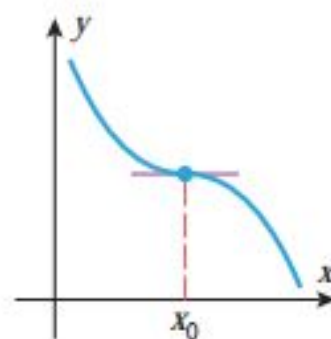
Critical point
Stationary point
Relative minimum



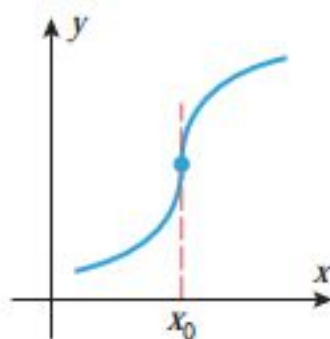
Critical point
Not a stationary point
Relative minimum



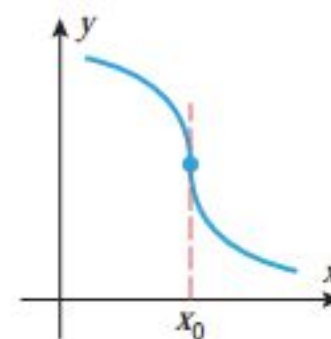
Critical point
Stationary point
Inflection point
Not a relative extremum



Critical point
Stationary point
Inflection point
Not a relative extremum



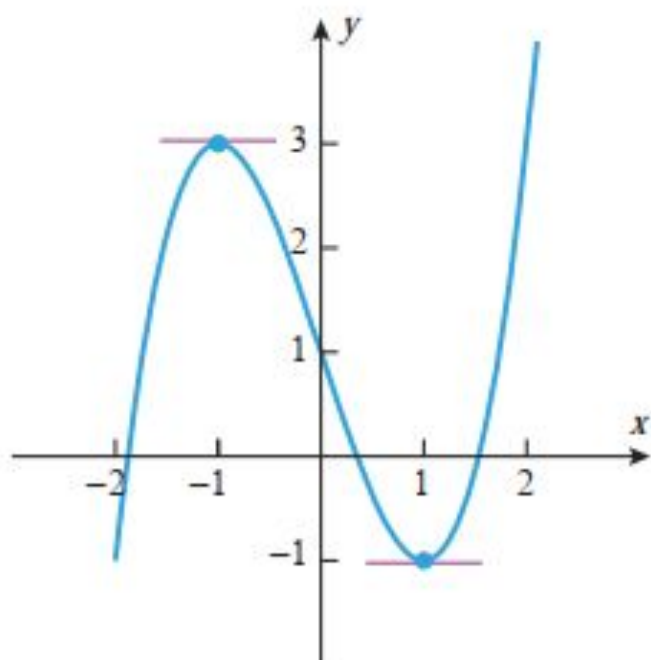
Critical point
Not a stationary point
Inflection point
Not a relative extremum



Critical point
Not a stationary point
Inflection point
Not a relative extremum

▲ Figure 4.2.6





$$y = x^3 - 3x + 1$$

▲ Figure 4.2.4

4.2.2 THEOREM Suppose that f is a function defined on an open interval containing the point x_0 . If f has a relative extremum at $x = x_0$, then $x = x_0$ is a critical point of f ; that is, either $f'(x_0) = 0$ or f is not differentiable at x_0 .

► **Example 2** Find all critical points of $f(x) = x^3 - 3x + 1$.

Solution. The function f , being a polynomial, is differentiable everywhere, so its critical points are all stationary points. To find these points we must solve the equation $f'(x) = 0$. Since

$$f'(x) = 3x^2 - 3 = 3(x + 1)(x - 1)$$

we conclude that the critical points occur at $x = -1$ and $x = 1$. This is consistent with the graph of f in Figure 4.2.4. ◀



1ST DERIVATIVE TEST

4.2.3 THEOREM (*First Derivative Test*) Suppose that f is continuous at a critical point x_0 .

- (a) If $f'(x) > 0$ on an open interval extending left from x_0 and $f'(x) < 0$ on an open interval extending right from x_0 , then f has a relative maximum at x_0 .
- (b) If $f'(x) < 0$ on an open interval extending left from x_0 and $f'(x) > 0$ on an open interval extending right from x_0 , then f has a relative minimum at x_0 .
- (c) If $f'(x)$ has the same sign on an open interval extending left from x_0 as it does on an open interval extending right from x_0 , then f does not have a relative extremum at x_0 .



$$f(x) = 3x^{5/3} - 15x^{2/3}$$

$$f'(x) = 3 \times \frac{5}{3} x^{5/3-1} - 15 \times \frac{2}{3} x^{2/3-1}$$

$$= 5x^{2/3} - 10x^{-1/3}$$

$$= 5x^{2/3} - \frac{10}{x^{1/3}}$$

$$= \frac{5x^{2/3}x^{1/3} - 10}{x^{1/3}} = \frac{5x - 10}{x^{1/3}} = \frac{5(x-2)}{x^{1/3}}$$

$$\text{Let } f'(x) = 0, \quad 0 = \frac{5(x-2)}{x^{1/3}}$$

$$\Rightarrow 5(x-2) = 0$$

$$\Rightarrow x = 2$$

$$\text{If } x=0, \quad f'(0) = \frac{5(0-2)}{0} = \text{undefined}$$

Thus, $x=0$ and $x=2$ are critical points and $x=2$ is a stationary point

$$f'(x) = \frac{5(x-2)}{x^{1/3}}$$

interval	$5(x-2)/x^{1/3}$	$f'(x)$
$x < 0$	$(-)/(-)$	$(+)$, increasing
$0 < x < 2$	$(-)/(+)$	$(-)$, decreasing
$x > 2$	$(+)/(+)$	$(+)$, increasing

sign changes (between $x < 0$ and $0 < x < 2$)

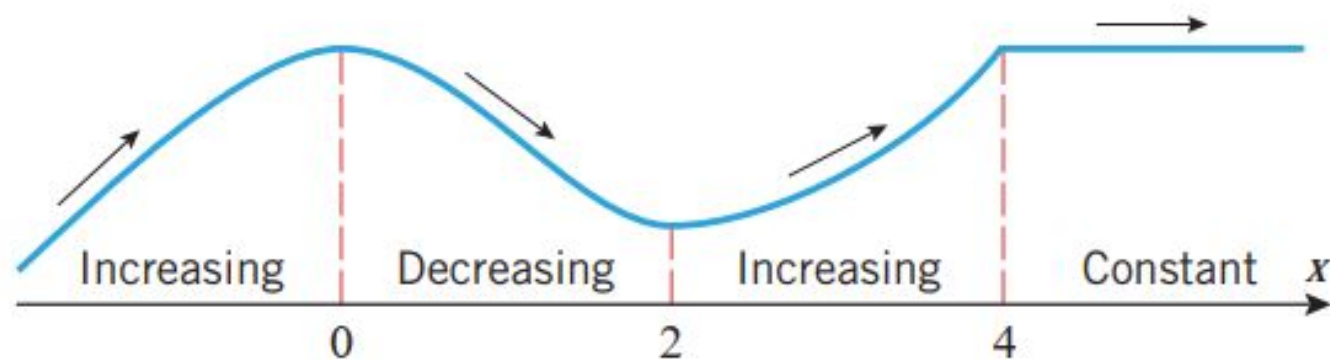
sign changes (between $0 < x < 2$ and $x > 2$)

→ The sign of f' changes from +ve to -ve at $x=0$, so there is a relative maximum at that point

→ The sign of f' changes from -ve to +ve at $x=2$, so there is a relative minimum at that point

■ INCREASING AND DECREASING FUNCTIONS

The terms *increasing*, *decreasing*, and *constant* are used to describe the behavior of a function as we travel left to right along its graph. For example, the function graphed in Figure 4.1.1 can be described as increasing to the left of $x = 0$, decreasing from $x = 0$ to $x = 2$, increasing from $x = 2$ to $x = 4$, and constant to the right of $x = 4$.



► Figure 4.1.1



2ND DERIVATIVE TEST

4.2.4 THEOREM (*Second Derivative Test*) Suppose that f is twice differentiable at the point x_0 .

- (a) If $f'(x_0) = 0$ and $f''(x_0) > 0$, then f has a relative minimum at x_0 .
- (b) If $f'(x_0) = 0$ and $f''(x_0) < 0$, then f has a relative maximum at x_0 .
- (c) If $f'(x_0) = 0$ and $f''(x_0) = 0$, then the test is inconclusive; that is, f may have a relative maximum, a relative minimum, or neither at x_0 .



► **Example 5** Find the relative extrema of $f(x) = 3x^5 - 5x^3$.

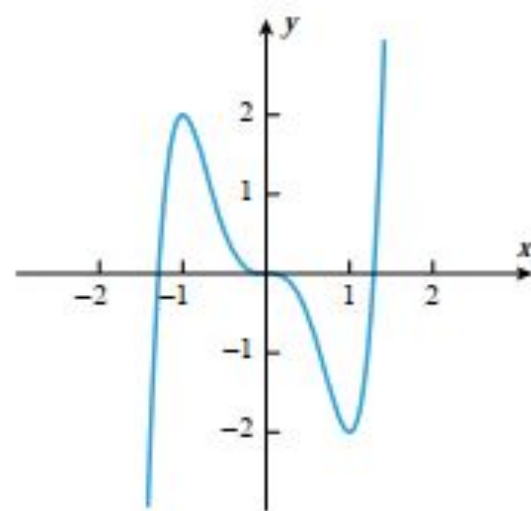
Solution. We have

$$f'(x) = 15x^4 - 15x^2 = 15x^2(x^2 - 1) = 15x^2(x + 1)(x - 1)$$

$$f''(x) = 60x^3 - 30x = 30x(2x^2 - 1)$$

Solving $f'(x) = 0$ yields the stationary points $x = 0$, $x = -1$, and $x = 1$. As shown in the following table, we can conclude from the second derivative test that f has a relative maximum at $x = -1$ and a relative minimum at $x = 1$.

STATIONARY POINT	$30x(2x^2 - 1)$	$f''(x)$	SECOND DERIVATIVE TEST
$x = -1$	-30	-	f has a relative maximum
$x = 0$	0	0	Inconclusive
$x = 1$	30	+	f has a relative minimum



$$y = 3x^5 - 5x^3$$

▲ Figure 4.2.8

The test is inconclusive at $x = 0$, so we will try the first derivative test at that point. A sign analysis of f' is given in the following table:

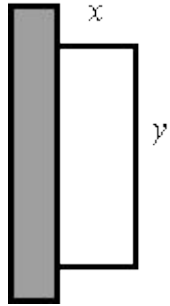
INTERVAL	$15x^2(x + 1)(x - 1)$	$f'(x)$
$-1 < x < 0$	(+)(+)(-)	-
$0 < x < 1$	(+)(+)(-)	-

Since there is no sign change in f' at $x = 0$, there is neither a relative maximum nor a relative minimum at that point. All of this is consistent with the graph of f shown in Figure 4.2.8. ◀



OPTIMIZATION

A rectangular storage area is to be constructed along the side of a tall building. A security fence is required along the remaining 3 sides of the area. What is the maximum area that can be enclosed with 800 m of fencing?



The area is $A = xy$

We know $2x + y = 800$ so $y = 800 - 2x$

So the area is $A = x(800 - 2x) = 800x - 2x^2$

To maximise the area, find when $\frac{dA}{dx} = 0$

$$\frac{dA}{dx} = 800 - 4x = 0$$

when

$$x = 200$$

Is it a maximum?

$$\frac{d^2A}{dx^2} = -4 < 0 \text{ for all } x$$

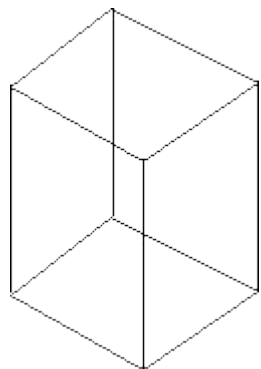
So it is a maximum.

So the maximum area occurs when $x = 200$, $y = 400$ and that area is:

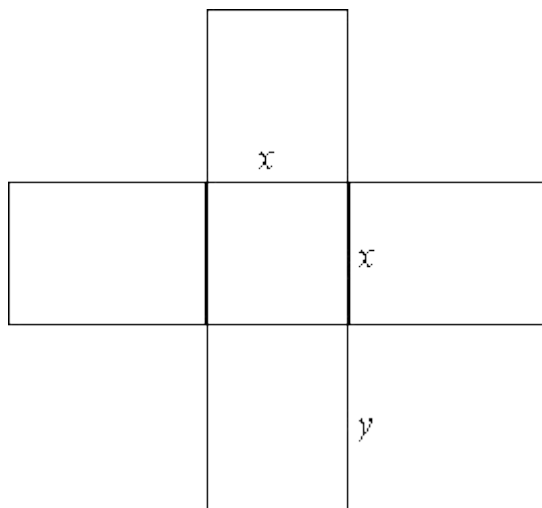
$$A = 200 \times 400 = 80\,000 \text{ m}^2 = 8 \text{ ha}$$



A box with a square base has no top. If 64 cm^2 of material is used, what is the maximum possible volume for the box?



The net for this box would be:



The volume of the box is $V = x^2y$

We are told that the surface area of the box is 64 cm^2 . The area of the base of the box is x^2 and the area of each side is xy , so the area of the base plus the area of the 4 sides is given by:

$$x^2 + 4xy = 64 \text{ cm}^2$$

Solving for y gives:

$$y = \frac{64 - x^2}{4x} = \frac{16}{x} - \frac{x}{4}$$

So the volume can be rewritten:

$$\begin{aligned} V &= x^2y \\ &= x^2 \left(\frac{16}{x} - \frac{x}{4} \right) \\ &= 16x - \frac{x^3}{4} \end{aligned}$$

Now

$$\frac{dV}{dx} = 16 - \frac{3x^2}{4}$$

and this is zero when

$$x = \pm \frac{8}{\sqrt{3}} \approx 4.62$$

(Note: The negative case has no practical meaning.)

Is it a maximum?

$$\frac{d^2V}{dx^2} = -\frac{3x}{2}$$

and this is negative when x is positive. So it is a MAX.

So the dimensions of the box are:

Base $4.62 \text{ cm} \times 4.62 \text{ cm}$ and sides 2.31 cm .

The maximum possible volume is

$$V = 4.62 \times 4.62 \times 2.31 \approx 49.3 \text{ cm}^3$$

Check: Area of material:

$$x^2 + 4xy = 21.3 + 4 \times 4.62 \times 2.31 = 64$$

Checks OK.



6.

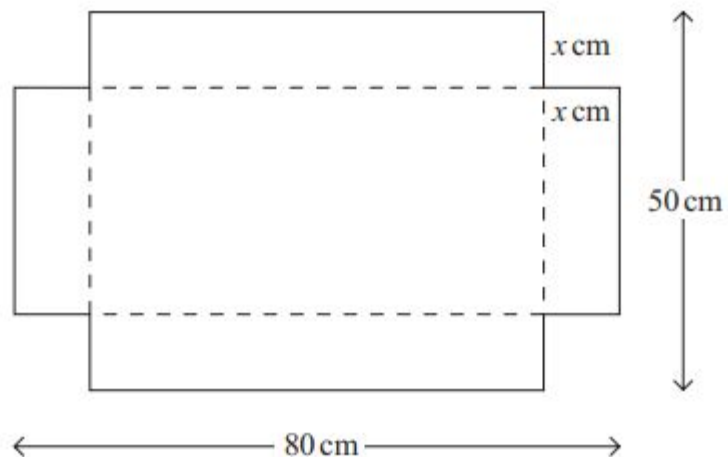


Figure 1

A rectangular sheet of card measures 80 cm by 50 cm. A square of side x cm is cut away from each corner, as shown in Figure 1. The card is folded along the dotted lines to form an open rectangular box of volume $V \text{ cm}^3$.

- (a) Show that $V = 4000x - 260x^2 + 4x^3$. (2)
- (b) Find the value of x for which V has its maximum value, justifying that this value of x gives the maximum value of V . (6)
- (c) Find the maximum value of V . (2)

